Mohammad Abu Shariha

HW5

Section 2.4

2.4-6

Given $n = 15$, $p = 0.75$. Then

$$X \sim \text{Binomial}(15, 0.75), \quad P(X = k) = \binom{15}{k}(0.75)^k(0.25)^{15-k}.$$

(a) $P(X \geq 10) = \sum_{k=10}^{15} P(X = k)$

$$P(X = 10) = \binom{15}{10}(0.75)^{10}(0.25)^5 = 3003(0.0563)(0.00098) = 0.1651,$$
$$P(X = 11) = \binom{15}{11}(0.75)^{11}(0.25)^4 = 1365(0.0422)(0.0039) = 0.2252,$$
$$P(X = 12) = \binom{15}{12}(0.75)^{12}(0.25)^3 = 455(0.0317)(0.0156) = 0.2252,$$
$$P(X = 13) = \binom{15}{13}(0.75)^{13}(0.25)^2 = 105(0.0238)(0.0625) = 0.1559,$$
$$P(X = 14) = \binom{15}{14}(0.75)^{14}(0.25)^1 = 15(0.0178)(0.25) = 0.0668,$$
$$P(X = 15) = (0.75)^{15} = 0.0134.$$

$$P(X \geq 10) = 0.8516.$$

(b) $P(X \leq 10) = 0.3135$.
(c) $P(X = 10) = 0.1651$.
(d) Mean, variance, and standard deviation:

$$\mu = np = 15(0.75) = 11.25, \quad \sigma^2 = np(1-p) = 15(0.75)(0.25) = 2.8125, \quad \sigma = \sqrt{2.8125} = 1.677.$$

2.4-8

Each valve opens with $p = 0.99$. Let X = number that open out of 4.

$$P(\text{none open}) = (0.01)^4 = 0.00000001.$$
$$P(\text{at least one}) = 1 - (0.01)^4 = 0.99999999.$$
$$P(\text{all 4 open}) = (0.99)^4 = 0.9606.$$

2.4-20

Find distribution, μ, σ^2, and $P(1 \le X \le 2)$.
(a) $M(t) = (0.3 + 0.7e^t)^5 \Rightarrow X \sim \text{Bin}(5, 0.7)$

$$\mu = 5(0.7) = 3.5, \quad \sigma^2 = 5(0.7)(0.3) = 1.05.$$
$$P(1 \le X \le 2) = \binom{5}{1}(0.7)(0.3)^4 + \binom{5}{2}(0.7)^2(0.3)^3 = 0.1607.$$

(b) $M(t) = \dfrac{0.3e^t}{1 - 0.7e^t} \rightarrow$ geometric with $p = 0.3$.

$$\mu = \frac{1}{p} = 3.33, \quad \sigma^2 = \frac{1-p}{p^2} = 7.78.$$
$$P(1 \le X \le 2) = 0.3 + 0.21 = 0.51.$$

(c) $M(t) = 0.45 + 0.55e^t \rightarrow \text{Bernoulli}(p = 0.55)$

$$\mu = 0.55, \quad \sigma^2 = 0.2475, \quad P(1 \le X \le 2) = 0.55.$$

(d) $M(t) = 0.3e^t + 0.4e^{2t} + 0.2e^{3t} + 0.1e^{4t}$

$$\mu = 1(0.3) + 2(0.4) + 3(0.2) + 4(0.1) = 2.1,$$
$$E[X^2] = 1^2(0.3) + 2^2(0.4) + 3^2(0.2) + 4^2(0.1) = 5.3,$$
$$\sigma^2 = 5.3 - (2.1)^2 = 0.89, \quad P(1 \le X \le 2) = 0.7.$$

(e) $M(t) = \sum_{x=1}^{10} 0.1e^{tx} \rightarrow$ uniform on $\{1, ..., 10\}$.

$$\mu = 5.5, \quad \sigma^2 = \frac{10^2 - 1}{12} = 8.25, \quad P(1 \le X \le 2) = 0.2.$$

Section 2.5

2.5-1

$N = 100$, $K = 5$ defectives, $n = 10$ sample.

$$P(\text{at least one defective}) = 1 - \frac{\binom{95}{10}}{\binom{100}{10}}.$$

$$\binom{95}{10} = 10{,}104{,}934{,}117{,}421, \quad \binom{100}{10} = 17{,}310{,}309{,}456{,}440.$$
$$P(\text{no defect}) = 0.5838, \quad P(\text{at least one}) = 1 - 0.5838 = 0.4162.$$

2.5-2

Eight men; 4 play north, 4 play south.
(a) Probability first four arrivals are all north:

$$P = \frac{4}{8} \cdot \frac{3}{7} \cdot \frac{2}{6} \cdot \frac{1}{5} = \frac{1}{70} = 0.0143.$$

(b) Probability first four all on same court:

$$P = 2 \times \frac{1}{70} = \frac{1}{35} = 0.0286.$$

2.5-9

$N = 50$, $K = 3$ defectives, $n = 10$.
(a) $P(X = 1) = \dfrac{\binom{3}{1}\binom{47}{9}}{\binom{50}{10}} = 0.398.$
(b) $P(X \leq 1) = P(X = 0) + P(X = 1)$

$$P(X = 0) = \frac{\binom{47}{10}}{\binom{50}{10}} = 0.5041, \quad P(X \leq 1) = 0.5041 + 0.398 = 0.902.$$